

Next-Gen AI Compiler

Agents On The Control Plane

Agents Own Search · Orchestration · Synthesis.
Compilers Own Lowering · Legality · Measure · Fallback.

Ask: Is The Hybrid Prediction (~2027-31) The Right Bet For Your Roadmap?

Expert Briefing · ~25-30 min + Q&A · Prediction-first

Agenda

- 
1. Trends · Keep Substrate
 2. Verdict + Architecture
 3. Roadmap Checkpoints
 4. Commercial Blockers + Gaps
 5. Technical Prediction (Techniques)
 6. Stance · Ask · Appendix

Trend backdrop — two stacks converging



Compilers for AI
mature · fragmented substrate

AI for compilers
rapid hybrid growth

Next-gen $\neq$ throw away LLVM/MLIR — make them **agent-addressable**.

Six active trends

A — Hybrid guidance

Free IR rewrite fragile
(mlirAgent < identity)
Constrain actions: hints /
advisory metadata / pass lists
AgentCompile · HintPilot · LLM Com-
piler

B — RL → agents

CompilerGym → tool agents
Heuristics as shippable C++
Workflow compile · freeze · place
Compiler-R1 · Magellan · FlowCompile

C — MLIR + Triton

PyTorch → Inductor → Triton
Also StableHLO → XLA / IREE
Peak often vendor libs / Tile
MLIR shared; product paths diverge

D — Kernel agents

KernelBench: correct *and* faster
One-shot often <20%; fusion hard
GEAK · KernelLLM · AgentCompile
Refine ≠ always faster

E — Verify in the loop

Unit/golden → numerical
→ Alive2-class local formal
Weak on GPU races / FP noise
Stacked oracle ladder for admit

F — Broader object

Mid-decode diagnosis
FMware: prompts / agents / knobs
Agents as compiler engineers
Compiler.next · Magellan · Claude C

Keep substrate, change control plane

Preserve

deterministic lowering
library kernels
build-CI regressability
local formal checks

Adopt

advisory search
multi-agent orchestration
heuristic synthesis
profile / verifier feedback

**Anti-pattern: replace opt/Inductor
with unconstrained LLM codegen**

Executive verdict

Agents reshape the **control plane**
more than they replace the **data plane**.

Control plane: search / orchestrate / synthesize

Data plane: lower / legality / measure / fallback

Compilers for AI
× AI for compilers

Hybrid LLM-compiler loops

4th job Tier A:
bring-up / codesign

Will not happen soon

- Unconstrained LLM replaces Inductor / opt
- Autonomous chip tape-out via compiler agents (C10)

Four agent jobs — architecture spine

(a) Online

propose →
measure
admit

CompileIQ / GEAK
ACCLAIM / HintPilot

(b) Offline

evolve heuristics
→ C++ / MLGO

Magellan
AlphaEvolve

(c) Eng.

oracle-gated
pull request
/ change

Archer

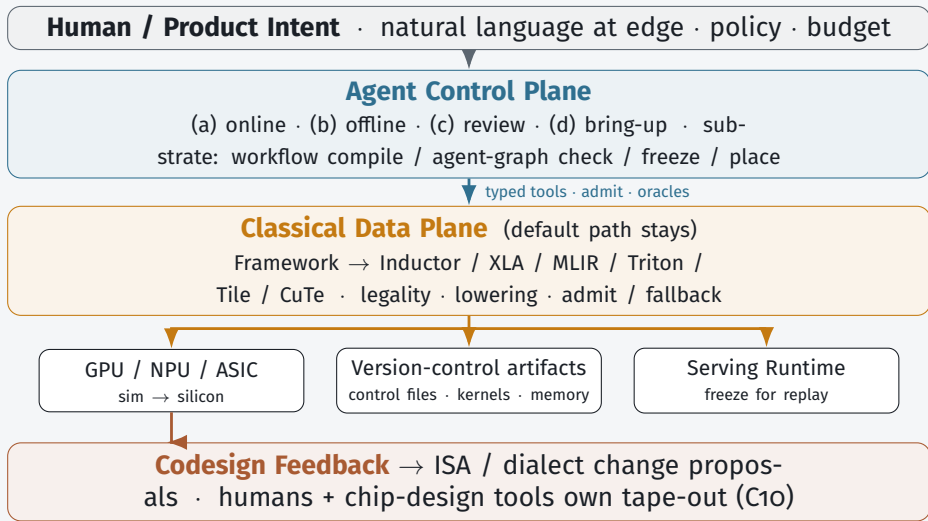
(d) Bring-up

coverage →
performance
sim + silicon

TritorX
KernelEvolve

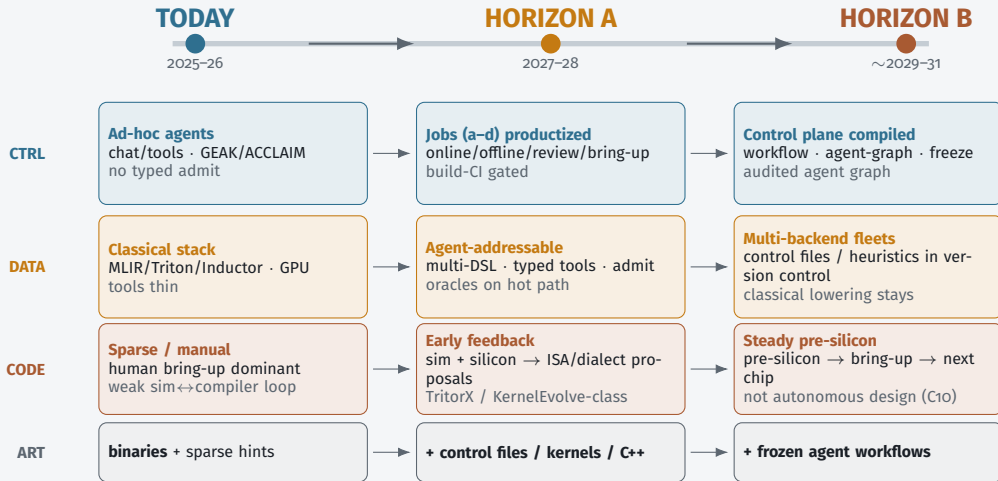
Control plane jobs sit *above* classical lowering — not instead of it.

Architecture — Target Stack



Invariant: LLM guides search — does not silently define unchecked executable behavior.

Architecture evolution — component changes



Components thicken left→right; data plane stays; agent graph becomes compiled & amortized.

What ships / does not by 2028

Predicted to ship

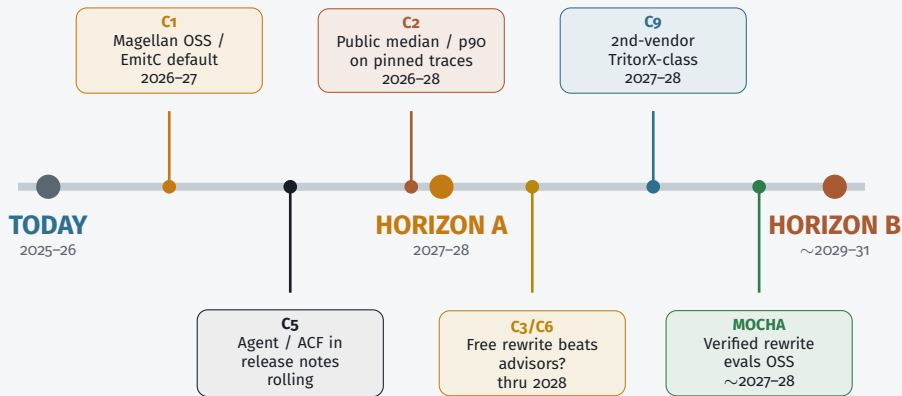
- Agent-addressable tool interfaces
- Hot-path specialize (not silent default-all)
- Magellan *and* MLGO both live
- Oracle pull-request review in serious orgs
- Coverage-first ASIC bring-up
- Triton-family primary; multi-DSL rising

Does not ship

- Unconstrained LLM replaces opt/Inductor
- One agent IR for all vendors
- Kernel agents uniformly beat eager on fusion ladders
- Autonomous microarch tape-out

Horizon A success = build-CI gated specialize + oracles + freeze artifacts

Roadmap Checkpoints — when the prediction changes



Falsifiable milestones — watch settlement signals; do not average disagreements.

Checkpoint C1 — heuristics vs neural advisors

A — Magellan path

evolve shippable C++ heuristics

Evolve shippable C++
(EVOLVE-BLOCK)

Offline coding agent
is the control-plane output

Product = evolutionary coding agent +
review

B — MLGO / EmitC path

MLGO = learned in-tree LLVM advisors

NN advisors stay in-tree

June 2026 plan of record:

internal inliner →

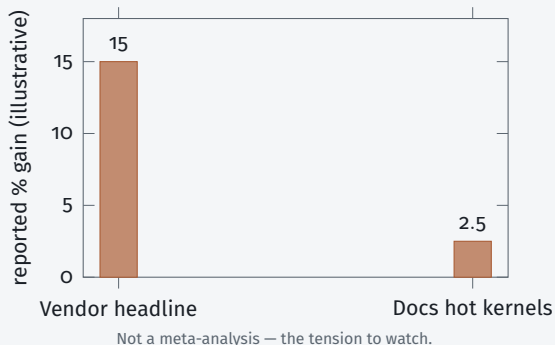
Android/Fuchsia →

Chrome multi-model

Agents train/feature NNs

Settlement: public Magellan llvm patches that dis-
place MLGO or EmitC-MLGO becomes customer default
Watch LLVM syncs + OpenEvolve recipes quarterly through 2027

Checkpoints C2 + C5 — gains & default path



C2 “Agents become default” needs **median (p50) build-C1 wins**, not best-kernel blogs.

p50 / p90 = 50th / 90th percentile of the gain distribution across builds.

C5 Online specialize (control files / hints) vs offline eng (Magellan/MLGO) — both may live; *which is the default flag?*

Settlement: pinned public traces + release notes listing agent / control-file workflows.

Checkpoints C₃ / C₆ / C₉ / C₁₀ — interface width & scope

C₃ — free rewrite vs advisory

Flip: shared suite where free rewrite wins

⇒ wide rewrite interface vs narrow control files / hints

C₉ — coverage vs peak

Flip: TritorX → KernelEvolve ladder public

⇒ SKU: coverage service target then performance target

C₆ — replace vs control plane

Flip: production default with *no* admit

⇒ agents replaced the compiler (we reject this)

C₁₀ — codesign vs auto tape-out

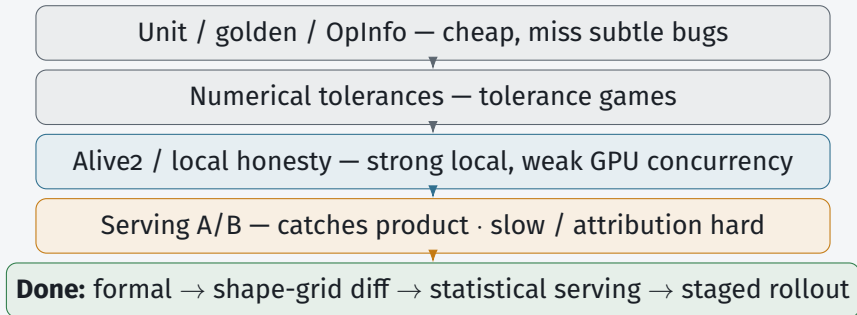
Flip: microarch primarily agent-proposed *and* compiler-oracle validated ⇒ enters chip-design tooling

Commercial blockers — top 5

- 1 Oracles strong enough for money
Fast-but-wrong · liability
- 2 Cost / replay / when agents run
Nondeterministic \$N compiles
- 3 Agent $\leftrightarrow$ compiler contract
Natural-language-only = demo
- 4 Distributional production evidence
No default-path A/B
- 5 Ownership · supply chain · human review
Unowned agent code in trusted base

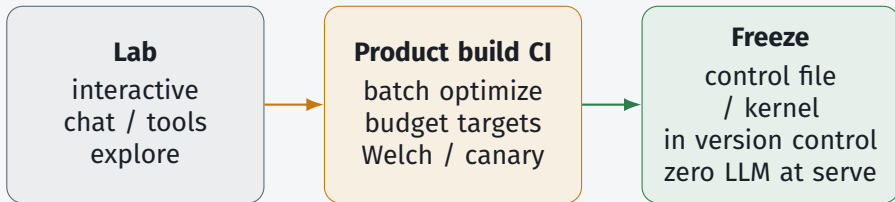
Each blocker gets one slide — productization gaps, not research wishlist.

Blocker 1 — Oracles for money



Room question: who owns false negatives when admit passes and production miscompiles?

Blocker 2 — Cost, replay, when may the agent run?



Cache key: (IR hash, hardware, compiler ver, agent policy) · budget \$/%gain

Falsifies commercially: “chat with compiler on every build” without spend/latency targets.

Blocker 3 — Agent $\leftrightarrow$ compiler contract

A. Natural language + pasted logs — demo only

B. Structured admit traces — build CI / bill of materials

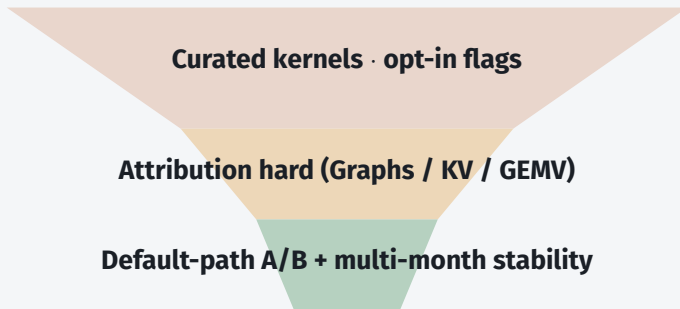
C. Typed tool interfaces — required action space

D. Hybrid — natural-language view over traces

```
admit_record = {graph_hash, hw_id, compiler_ver, action[], oracle[],  
artifact_digest, policy_id}
```

Lean: **C + B**; Natural language is a view, not source of truth. Advanced Control File = portable compiler knobs.

Blocker 4 — Distributional production evidence



Marketing bar: median + 90th-percentile gains + cost-to-compile
· persistent serving throughput, not single-kernel headlines

Blocker 5 — Ownership, security, human review

Trusted-base risk

agent kernels /
heuristics enter
trusted base

Ownership

CODEOWNERS
signed provenance
sandbox

Human-review capacity

agents $\uparrow$ drafts
reviewers =
bottleneck

Lean: human CODEOWNER + signed admit + sand-
box · oracle auto-merge only for **narrow** action classes
Generic forge AI $\neq$ compiler-oracle review (C7)

Gap map — what blocks the prediction, not a wishlist

HIGH

4.1 prod evidence

4.2 correctness

4.3 cost/replay

4.4 cross-stack

4.10 benchmarks

MED-HIGH

4.5 hardware-native interfaces

4.7 training data

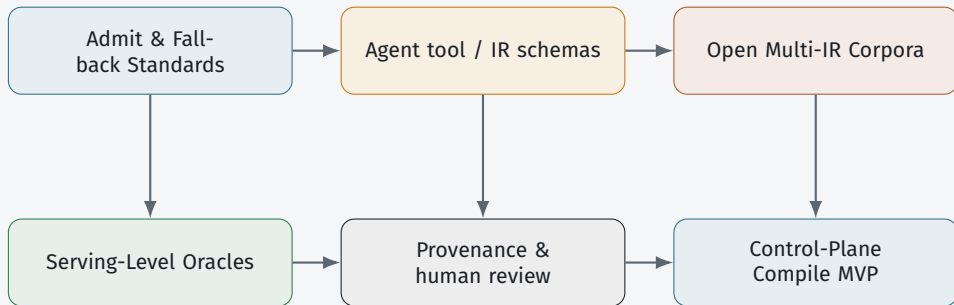
MEDIUM

4.6 workflow compile

4.8 human-review process

4.9 security

Cross-Cutting Research Agenda



Near-term research themes → next: technique map T1–T10 (exists / missing / unlocks).

Technical Prediction — Accelerate The Roadmap

To settle checkpoints and ship Horizon A,
enhance techniques *inside and outside* the compiler (T₁–T₁₀).

Within (T₁–T₅)

- T1 Typed agent ↔ compiler interfaces
- T2 Admit / fallback machinery
- T3 Control files, hints, fingerprints + replay
- T4 Heuristic hooks & in-tree advisors
- T5 Dialect / ISA feedback sinks

Outside (T₆–T₁₀)

- T6 Serving-level oracles & production A/B
- T7 Open multi-IR corpora (+ negatives)
- T8 Unified benchmark ladder
- T9 Provenance, ownership, HITL process
- T10 Agent-workflow compile / freeze / place

Enhancing only opt/Inductor/Triton internals is not enough. Next: exists · missing today · checkpoint unlocks.

Technical Prediction — Within The Compiler (T_I–T₅)

T₁ Typed agent ↔ compiler interfaces **Exists:** CompileIQ, ACCLAIM/HintPilot, mlirAgent MCP, mlir-opt-repl, FlashInfer Trace. **Missing:** portable schemas across MLIR·Triton·Tile·StableHLO. **Unlocks:** C3, C5, C6.

T₂ Admit / fallback machinery **Exists:** AgentCompile checks, Archer oracles, TritonRL verifiers, FlashInfer-Bench gates, VibeServe judges. **Missing:** shared admit *product* + trusted deterministic fallback. **Unlocks:** C6 hybrid; money-grade shipping.

T₃ Control files, hints, fingerprints + replay **Exists:** CompileIQ ACFs; FlashInfer Trace + apply(). **Missing:** content-addressed keys (IR×HW×compiler×policy); golden replay on model upgrade. **Unlocks:** C2, C5.

T₄ Heuristic hooks & in-tree advisors **Exists:** Magellan / AlphaEvolve C++; MLGO advisors; EmitC-MLGO PoR advancing. **Missing:** settled *default* on named shipping apps. **Unlocks:** C1; Horizon A job (b).

T₅ Dialect / ISA feedback sinks **Exists:** TritonX / KernelEvolve / Ascend diagnosis (sim+silicon → IR stress). **Missing:** first-class compiler surfaces for dialect/ISA *change proposals* (humans+EDA still own tape-out). **Unlocks:** C9, C10.

Technical Prediction — Outside The Compiler (T6–T10)

T6 Serving-level oracles & production A/B **Exists:** unit/golden/Alive2; **FlashInfer-Bench** serving traces + `apply()`; VibeServe judges. **Missing:** whole-program/GPU-race/FP contracts; multi-month *default-path* A/B. **Unlocks:** C2.

T7 Open multi-IR corpora (+ negatives) **Exists:** Meta LLM Compiler; **KernelBook** → KernelLLM/TritonRL; DR Triton synthetic 100k. **Missing:** versioned MLIR/Tile/StableHLO + failed/miscompile/slow-correct negatives. **Unlocks:** selectors beyond LLVM-centric models.

T8 Unified benchmark ladder **Exists:** KernelBench(-X); FlashInfer-Bench closes a *serving-kernel* rung. **Missing:** shared IR→kernel→fused→serving ladder with cost-to-compile. **Unlocks:** honest C2/C9.

T9 Provenance, ownership, HITL process **Exists:** Magellan reviewable C++; Archer oracle review; sparse SBOM talk. **Missing:** CODEOWNERS + signed admit records + sandbox for untrusted proposals. **Unlocks:** trusted-base shipping; C7.

T10 Agent-workflow compile / freeze / place **Exists:** FlowCompile, Auto, AgentFlow, hetero place; VibeServe (early evidence *now*). **Missing:** shared agent-graph IR + fail-closed CI (productization → Horizon B). **Unlocks:** Horizon B control-plane compile.

Technical Prediction — What Each Checkpoint Needs

C1 ← **T4** Magellan-class synthesis *or* EmitC-MLGO customer default on named apps.

C2 ← **T3+T6+T8** replayable artifacts + serving oracles + ladder; FlashInfer-Bench = pressure, not settlement.

C5 ← **T1+T3** typed interfaces + freeze-before-serve named in release notes.

C3 / C6 ← **T1+T2** constrained tools + admit/fallback (lean: hybrid advisory until free rewrite proves out).

C9 ← **T5+T8** coverage→perf agents + dialect sinks; need second-vendor TritonX-class path.

C10 ← **T5** failure modes → ISA/dialect *proposals only*; humans + chip-design tools own tape-out.

Technical Prediction — Critical Missing Parts Now

Highest-leverage bets to accelerate Horizon A

1. Money-grade oracle stack (T2+T6)

local formal → shape-grid diff
→ serving statistics → staged rollout
Without this, agents stay demos

2. Replayable artifact contract (T3)

control files / kernels / heuristics
with cache keys + golden replay
Without this, CI rejects the loop

3. Portable agent compile interface (T1)

summaries · actions · admit records
across MLIR / Triton / Tile / StableHLO
Without this, every stack re-glues

4. Open ladder + multi-IR data (T7+T8)

correctness × speed × \$/compile
plus negative (failed) examples
Without this, claims stay incomparable

Survey lean: T1–T5 must ship as product surfaces; T6–T10 are equally first-class — mostly outside classical lowering.

Org Adoption Questions

1. Online (CompileIQ) vs Offline (Magellan) — which matches your release model?
2. Oracle stack: Alive2 / golden / serving A/B — who owns false negatives?
3. Agent contract surface: LLVM / MLIR / Triton / StableHLO / Tile?
4. How do you cache/regress traces across compiler *and* model upgrades?
5. Max \$/build for a median X% win? Named maintainer per admitted artifact?

Working stance until conflicts settle

- 1 Hybrid: agents search; compilers lower (not replace)
- 2 Magellan || MLGO — parallel bets (C1)
- 3 Discount single-number speedups (C2)
- 4 Constrained actions + strong oracles (C3)
- 5 Demote generic SCM AI (C7)
- 6 Coverage→performance ladder; no autonomous chip design (C9/C10)

Commercial checklist — handout

Architecture

typed tools + admit
traces
version control $\gg$ dense
memory $\gg$ scratchpad
state-machine plan for
service targets
freeze before default-on

Trust

layered oracles
CODEOWNERS + prove-
nance
joint version pins
A/B before “prod default”

Business

sell regressable artifacts
+ build-CI quota
customer owns outputs
coverage then perfor-
mance service target
token budget per %gain

Discussion

Thank you — hybrid bet stays the ask

1. Which checkpoint flips your investment first — C1, C2, or C9?
2. Harder commercial bar: **oracle strength** or **replay/freeze**?
3. Building job (a), (b), or (d) first — what do you refuse online?

Ask again: is the hybrid bet right for your roadmap? Watch settlement signals — do not average disagreements.

Appendix

Evidence Store Snapshot

Publications digests · commercial product signals · open repositories.
Tiered for the agentic-compiler prediction — not a full catalog.

Appendix — evidence map

Publications

~115 digests
one file per source
papers · blogs · talks
forge pages · minutes
★ = prediction-critical

Products

Commercial SKUs as
prediction signals
Tier A shapes the future
Tier B = data-plane de-
faults
not a full SKU list

Repositories

GitHub / Gerrit / forge
artifacts tiered A/B/C
A = agents reshape com-
pile
C = generic forge AI only
not an exhaustive forge
map

Use ★ digests + Tier A products/repos when a checkpoint or blocker is contested.

Appendix — Tier A commercial signals

Company offerings that shape what ships — roles: agent loop / kernel surface / cloud agent

Google / DeepMind — AlphaEvolve Cloud (GA); Magellan + MLGO (offline heuristics *and* neural advisors)

NVIDIA — CompileIQ (+ agent-skills); CUDA Tile / Tile IR; TensorRT-LLM agent skills

AMD — GEAk (multi-agent Triton / Instinct kernel optimization)

Meta — LLM Compiler / KernelLLM; TritonX + KernelEvolve (ASIC bring-up); Helion

FlashInfer / NVIDIA · UW · CMU — FlashInfer-Bench (serving-trace ladder + apply() into SGLang/vLLM)

Tier B baselines: TensorRT-LLM · Inductor · XLA/StableHLO · FlashInfer kernels · Modular MAX · OpenVINO · Neuron · Hexagon-MLIR

Appendix — Tier A open repositories

Offline / review / heuristics

OpenEvolve (Magellan-class loop)
HeuriGym (heuristic bench)
Archer (oracle pull-request review)
Compiler-R1 · HintPilot
mlirAgent (negative free-rewrite)

Online / kernels / bring-up

ACCLAIM · CompileIQ
GEAK · KernelAgent
KernelBench / KernelBench-X
FlashInfer-Bench · AutoKernel
Helion · TritonX / KernelEvolve

How to read tiers: A = agents change heuristics/kernels/knobs/review with domain oracles. B = data-plane hosts. C = generic forge AI (demoted for prediction).

Appendix — prediction-critical digests (★)

Highest-signal sources for the hybrid prediction — mechanisms, not headlines

Vision / funded: Compiler 2.0 Ken Kennedy plenary (MIT) · DARPA MOCHA / Aarno Compiler 2.0

Offline / online jobs: Magellan (+ LLVM slides) · ACCLAIM (+ GitHub) · EmitC-MLGO June 2026 plan-of-record minutes

Bring-up / codesign: TritorX · KernelEvolve · Ascend hierarchical diagnosis · KForge

Kernels / ladder / data: CompileIQ agent-skills · FlashInfer-Bench ★ · KernelBook ★ · AutoKernel · CuTeGen · Auto · Flow-Compile

Full digest index: one file per source under the publications store (~115 entries). Technique map: SURVEY §5.8 T1–T10.

Appendix — publication groups

17 GPU kernels & inference
13 Agentic & RL compilers
13 Source control & review

10 Classic DL compilers
10 Company infra
9 Forums & workshops

8 Surveys & vision
8 Foundation LLMs
6 MLGO & RL gyms

5 HW codesign & bring-up
4 Commercial products
4 Control-plane substrate

How digests are used

Mechanism first; demote generic forge AI.

When sources disagree, keep both sides — do not average.

Update prediction only when Tier A / ★ evidence moves.